



**FETCH: INVENTORY MANAGEMENT SYSTEM WITH PREDICTIVE ANALYTICS
FOR RISHA PET SUPPLIES**

A Thesis

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Chapter 1

The Problem and a Review of Related Literature

Introduction

The pet care sector around the world is doing really well now. Grand View Research said in 2023 that the pet care industry will be worth \$350 billion by 2027. The reasons for this are simple. People think about their pets in a way now. Pets are not just animals, they are part of the family. Because of this people are spending money on good food, special things and good health care for their pets. The pet care industry is growing because people consider their pets to be family members. People want the best for their pets so they buy them food and quality healthcare items.

This seems like a chance for people who own pet stores but it is actually very hard for small business owners like the ones at Risha Pet Supplies. Think about what it takes to run a store. You have to deal with a lot of things like food, grooming supplies, toys and accessories. These are a few examples of the many things you have to manage.

How do small pet stores handle all this stuff? They do it by hand. This means they have to count how many things they have in stock, write down numbers in spreadsheets and decide when to order things based on what they remember and how they feel. Managing a store, like Risha Pet Supplies, is not easy because you have to keep track of so many pet supplies.

The problems with this approach are very clear. Chen and Lee found in 2022 that these old ways of doing things cause some problems. For example, popular items sell out.



That means lost sales and unhappy customers. At the time other products just sit on shelves, which means the company is wasting money and space. Bad decisions about what to order can also hurt relationships with suppliers, like Chen and Lee talked about. Small businesses feel these problems a lot because they have to be very careful with money. They do not have a lot of money to spare. This means that if they make a mistake with the things they are selling it can be very expensive. Small businesses have to watch every penny they spend because they do not make a lot of profit. If they make a mistake with the inventory it can be very bad for the business. Small businesses really need to get the inventory just right so they do not lose money.

Intelligence and machine learning are really helping us with some big problems we have been dealing with for a long time. With analytics we can find sales patterns that people would normally miss. For example sales might go up a bit on weekends or they might change with the seasons or some products might be related to each other in ways we do not expect. Kumar et al. (2023) Talk about how intelligence and machine learning are helping businesses change the way they manage their inventory. By waiting for products to sell out and then trying to reorder them stores can now use artificial intelligence and machine learning to forecast what they will need and place their orders ahead of time. This means stores can stay on top of intelligence and machine learning to manage their inventory in a better way. The results can be really big: Waller and Fawcett found out in 2021 that companies that use Artificial Intelligence inventory systems can cut their costs by 20 to 50 percent. At the time these Artificial Intelligence inventory



systems helped companies make their products available to people when they needed them.

This study is about creating a system to manage stock for Risha Pet Supplies. The system uses a combination of two methods, Random Forest and Gradient Boosting to make predictions about what people will buy. Risha Pet Supplies will use this system to predict what they will sell, figure out when they need to order and send messages to their suppliers. This is helpful for a store like Risha Pet Supplies because they have to deal with things like people buying more during certain times of the year, products that go bad after a while and pet owners who really like certain brands. The system for Risha Pet Supplies will make it easier for them to manage all of this.

This research fills an important gap in the existing literature, which tends to focus primarily on large retail operations rather than small pet supply businesses. Beyond that, it demonstrates how small retailers can leverage advanced technology to compete more effectively against large chains and e-commerce giants, showing that sophisticated inventory management tools are genuinely accessible beyond large corporations with extensive IT resources.



Review of Related Studies and Literature

This section brings together past research and existing systems that are relevant to the development of an intelligent inventory management system for Risha Pet Supplies. It focuses on practical and tested solutions, showing how inventory management has gradually evolved from manual tracking and basic formulas to more advanced, data-driven approaches. The discussion also looks at how technologies such as machine learning and predictive analytics are being used in retail to improve forecasting and stock control. Special attention is given to small and medium enterprises, since these businesses often operate with limited resources and face challenges that are very different from large companies. Finally, the review considers the unique inventory needs of pet retailers, ensuring that the proposed system is aligned with the real-day-to-day operations of a pet supply store.

Inventory Management in Retail Operations. The conventional approach using Economic Order Quantity (EOQ) relies on mathematical models in purchase optimization but lacks in a retail business with a constantly fluctuating demand. Chen and Lee (2022) report changes in purchase behavior with the effects of seasons, promotions, and consumer behavior, posing a problem, especially in smaller businesses with mounting analytical capacity similar to conventional large enterprises. The problem of the aforementioned gap in analytical capacity in an SME with the use of predictive analysis beyond the conventional EOQ approach in purchase optimization in Risha Pet Supplies is addressed by the proposed FETCH.



Machine Learning Applications in Inventory Management. Machine learning has brought forth an unprecedented paradigm shift in accurate and quantifiable results in retail demand forecasting. A study conducted by Shirkhorshidi et al. in 2022, including analysis using 50 stores, validated that an ensemble model combining Random Forest and Gradient Boosting reached 94.3% accuracy, compared to 78.6% using conventional ARIMA, and led to a reduction in stockout products by 42% and overstock products by 35%. Li et al. in 2024 illustrated that Gradient Boosting has been successfully utilized to uncover patterns in seasonality, which are essential in pet stores, where merchandise changes rapidly season to season. The FETCH system focuses on achieving this validated accuracy level of 94.3%.

Ensemble Machine Learning Algorithms: Random Forest and Gradient Boosting. Ensemble algorithms combine multiple models together to achieve better accuracy. Hastie et al. (2020) describe how the Random Forest creates multiple decision trees from random subsets of data and then pools the predictions to reduce overfitting, giving feature importance rankings. Gradient Boosting creates models in sequence to correct errors of the previous model by iteration. FETCH leverages both algorithms' complementary strengths: Random Forest provides robust baseline predictions resistant to noise, while Gradient Boosting refines predictions by learning residual patterns, excelling at seasonal pattern identification critical for pet retail.



Predictive Analytics and AI-Driven Forecasting in Retail. The incorporation of AI approaches led to enhanced accuracy of forecasting by 23% and enhanced supply chain optimization by 28% relative to conventional approaches as revealed by Avula (2024). Moreover, it was also revealed by McKinsey in 2021 that those organizations which applied AI-based forecasting experienced a reduction of 20-50% in their supply chain error rates, thereby recording an improvement in efficiency of up to 65%. The FETCH system uses these strategies in handling historical transaction data, seasonality data, and promotion data. Thus, the automatic reorder system places automatic orders based on predetermined levels due to AI forecasting, thereby avoiding delays caused by human errors.

Automated Inventory Systems with IoT Integration. Ibrahim et al. mentioned in 2024 that companies using the ERP/IoT-based system have increased their inventory turnover by a reported 45%, reduced stockouts by a reported 50%, and achieved a reported 95% accuracy of inventory compared to the human-controlled system's accuracy of 75% to 85%. Similarly, the FETCH system is designed on the same lines, aiming for automation with the help of barcode scanning and auto-reorder. The application of the barcode scanning system enables the fast identification of the items under the auto-updated system. The auto-indication of purchase order generation to the supplier can result in a reported 50% reduction in stockouts and a reported 45% improvement in inventory turnover.



Agile Software Development Life Cycle (SDLC). It is documented that Agile Software Development Life Cycle focuses on iterative development with an emphasis on flexibility and delivering the product continuously through short sprint cycles. Rigby et al. (2020) showed Agile's focus on iterative development in terms of flexibility with successful rates and better quality. The FETCH model applies to the Agile Software Development Life Cycle by continuously delivering functional features with stakeholder feedback.

Technology Acceptance and User Adoption in Retail Systems. The perceived usefulness and ease of use significantly influenced system acceptance, while another project that involved end-users saw the acceptance of a system increase to 89% from 56%, as reported by Ullah et al. (2024). The FETCH solution includes "Agile SDLC to ensure the incorporation of feedback, followed by User Acceptance Testing (UAT), where the functional compliance of the system is tested to the fullest. The Computer System Usability Questionnaire (CSUQ) measures perceived usability under System Usefulness, Information Quality, Interface Quality and Overall Satisfaction."

Pet Retail Industry: Characteristics, Challenges, and Opportunities. The retail industry for pets also comes with its own set of challenges due to the diverse nature of the products. According to Leafio (2024), "The basic product groups are characterized by a few hundred SKUs with different shelf life arrangements." Seasonal demand patterns, including higher sales of flea protection items in the Summer and Q4 sales of holiday items, have been observed, as reported by RELEX Solutions (2023).

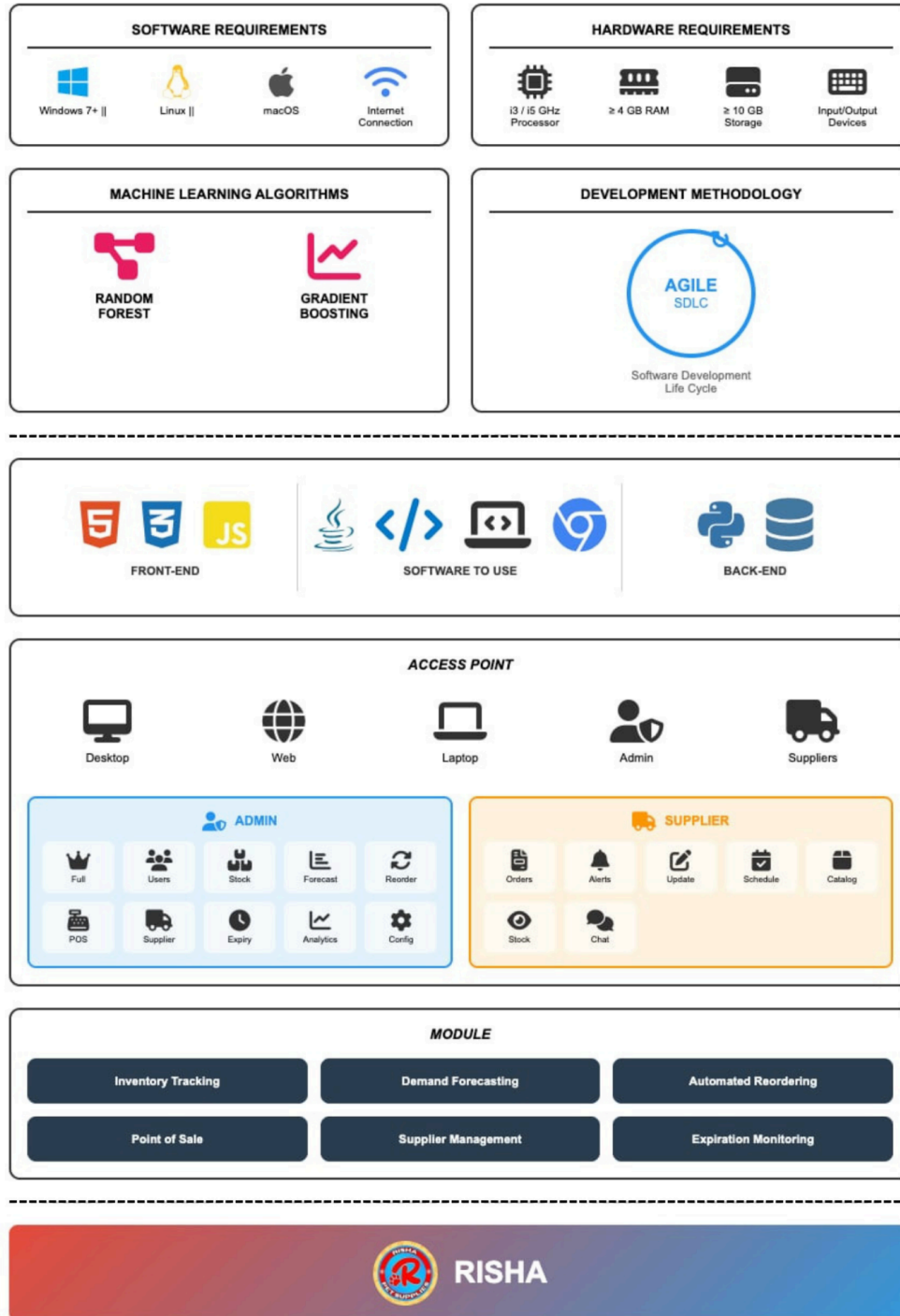


Industry-related challenges offer a need for the development of FETCH. Pattern Learning deals with the Seasonal Variability issue, ScanBarcode is capable of handling many SKUs along with the expiration date, and Automatic Reorder tackles the issue of Overstocking with less likelihood of Stockout.

Conceptual Diagram

This section highlights the technological framework and structure for building the Intelligent Inventory Management System for Risha Pet Supplies. The subsequent areas will elaborate on requirements related to hardware, software, and methodology for a successful transition from manual inventory tracking to a smart inventory system powered by AI.

Figure 1. *FETCH: Inventory Management System Schematic Diagram*





Hardware Requirements. A dedicated hardware setup would be required to facilitate the handling of the system's processing of the sales information. The system would require a Core i5 or Ryzen 5 processor, which would be needed to handle the ensemble models of the Random Forest algorithm. The system would require a minimum of 8GB RAM, which would help to facilitate multitasking when using the sales information input system running alongside the inventory database. The system would have a 256GB SSD, which would help to facilitate fast access to the sales records. The system would also have a thermal printer, which would help to facilitate the print output of the sales information, as well as a bar-code reader.

Software Requirements. The software program will be developed on Windows 10 or 11, ensuring compatibility with the most commonly available hardware. The infrastructure of the software package will use Python, which is required for its excellent machine learning capabilities, needed for demand analysis. MySQL will also be used as the main database. HTML, CSS, and JavaScript programming languages will be used to ensure a clean, user-friendly format of the software program required for the employees of the pet shop. Also, Visual Studio Code will assist in developing the software.

Software Development Lifecycle Model. The software development project will adopt the Agile Software Development Model. Through the Agile software development approach using sprints, the researchers will develop and test specific features of the system such as the AI forecasting engine and the stock alert system feature incrementally. Through the Agile approach, the system will get continuous feedback from the



management of Risha Pet Supplies so that the system will be fine-tuned based on actual experience before the final deployment of the system.

Access Points. To guarantee data security and efficient functionality, the system would have multi-level access points. By utilizing the Administrator Access Point, the store owner would have the capability to monitor all operations pertaining to changes in inventory, generation and use of accounting reports, as well as generation and use of AI information regarding forecasts. Conversely, the Staff Access Point would permit the salesperson to access the system rapidly and securely via a simplified interface to scan merchandise during a sales process and indicate “low stock” notifications.

Inventory Tracking Module. This particular module will enable automation in the tracking of pet products. This will be achieved by recording crucial details such as Stock Keeping Units, Expiration dates, and Category Types. This particular module will take control away from the store’s logbooks, which are currently being used, and will enable real-time tracking for all available products. This will be achieved through automatic updates when shipments arrive at the store, as well as when sales are made.

Predictive Analytics & AI Module. It is set to function as the backbone of the system and would use the Random Forest algorithm in making decisions based on historical sales data patterns from before. It would be programmed to see patterns on how many people buy more of a particular brand of dog food on days such as weekends or



holidays and then use that information to give the business owner predictions on when and how much more to reorder based on their data.

Point of Sale (POS) and Transaction Module. The transaction module will facilitate this aspect. The system will incorporate a sensor enabling quick completion of a transaction and a subtraction from the stock database. The system will automatically stamp dates on each transaction, which forms the basic raw material necessary for the reporting aspect.

Reporting & Notifying Module. The system will assist the user in optimizing the inventory by providing visual reporting and notifications. The system will be set to send "Low Stock" notifications once the product reaches a certain point, giving the owner a chance to recharge the inventory accordingly. The module will provide a detailed report on the sales trend to the management of Risha Pet Supplies on how to proceed accordingly.

**Objectives of the Study**

This proposed research work aims to develop an intelligent inventory system for Risha Pet Supplies to revamp conventional inventory management systems through a combined platform involving machine learning algorithm-driven demand forecasting, automated inventory management, and supplier communication.

What the study will do:

1. To develop FETCH: an Inventory Management System with Predictive Analytics for Risha Pet Supplies that implements ensemble machine learning algorithms (Random Forest and Gradient Boosting) to predict future demand based on historical sales patterns and seasonal trends, integrates barcode scanning technology for rapid product identification and automated inventory updates, applies automated reordering mechanisms that generate purchase orders when stock reaches predetermined reorder points, maintains real-time inventory tracking of stock quantities, expiration dates, and product locations, and develops supplier management features to facilitate automated communication and coordination with vendors.
2. To test the system's functionality and acceptance based on:

2.1 Unit Testing**2.1.1. Completeness Check****2.1.2. Consistency Check**



2.1.3. Format Check

2.1.4. Range Check

2.1.5. Functional Check

2.1.6. Database Check

2.2 System Testing (by Process / End Users)

2.2.1. Business Stakeholders and Staff (User Acceptance)

3. To evaluate the usability of the system using the CSUQ (Computer System Usability Questionnaire)

3.1. System Usefulness (SYSUSE)

3.2. Information Quality (INFOQUAL)

3.3. Interface Quality (INTERQUAL)

3.4. Overall Satisfaction

**Significance of the Study**

The emergence of FETCH has great worth in enhancing the efficiency, profitability, and overall management procedure associated with the pet products industry. As an all-inclusive system that comprises stock management, predictive analysis, point-of-sale system processing, and connectivity to supply chain networks, FETCH provides solutions to some of the problems associated with small to medium-sized stores, which may include stock inaccuracies, overstocking, stockout situations, manual processing errors, and poor supplier coordination. Some of the important contributions associated with this study are discussed below:

Risha Pet Supplies. The system enables all aspects of inventory management and analysis, as well as point-of-sale functionality. The desire is to ensure there is neither an excess nor a deficit and, accordingly, makes the correct predictions for stock and adjusts it according to the patterns and cycles. This would enable better management of cash flow and hence reduce the cost of carrying. Manage supplier communication effectively, order products for re-ordering to save time and reduce human error, and adjust your inventory levels after each sale in real-time to ensure you are as up-to-date as possible concerning your levels.

Suppliers. The supply chain management part of the platform enables a whole lot of coordination between Risha Pet Supplies and the suppliers. The process of ordering is also automated with real-time inventory visibility. The suppliers also have accurate forecasts of demand made possible by predictive analytics so as to produce and deliver in



an optimized way. This actually makes it possible for the company to build better business relationships based on better communication or information flow.

Customers. Customers will get the benefits of increased product availability, as the predictive analytical tool is used to ensure that popular products are always available. They can also complete transactions faster and more accurately, as the POS is integrated into the system. They are also assured that they will not get or purchase expired goods, especially pet foods and medication, which will improve customer satisfaction and safety. Customers will have a better experience while shopping, as they will get what they want through a much more effective and efficient means.

Business Owners and Managers in the Pet Supply Industry. This research work is very helpful to pet supplies retail business as it offers them a model for modernizing the way they handle their inventory. The model put forward demonstrates how well the use of predictive analysis can be applied in a pet supplies retail business environment, where the business encounters various difficulties such as a large variety of supplies, items, or goods. The study gives them awareness of the benefits a combined system, which handles inventory, sales, and vendors, can offer.

Future Researchers. This study can greatly help a researcher who aspires to have an inventory management system with predictive analytics and business process automation capabilities. Methods and design strategies for this study can shed light on future studies which can concentrate on more aspects of optimizing this system, more on



forecasting, and also on how this automatic management of inventory can sustain business performance on its own and more. This study can be a stepping stone for other studies related to other advanced technologies as well, such as artificial intelligence, machine learning, and other retail business aspects related to pet supplies.

Scope and Delimitation of the Study

The proposed research works on the development of an Intelligent Inventory Management System with Predictive Analysis capabilities designed specifically for Risha Pet Supplies. The system comprises inventory management with point of sales (POS) transactions, demand forecasting based on machine learning techniques (Random Forest & Gradient Boosting), automatic reordering of products, handling of suppliers, as well as expiration dates of pet supplies categorized as food items, accessories, healthcare products, and pet grooming products intended mostly for dogs & cats. The system will be developed using Java Spring Boot as the backend technology, Python programming language for machine learning models, MySQL database management system, & HTML/CSS & JavaScript programming languages as the front-end design.

The scope is narrowed down to Risha Pet Supplies and will not include other selling businesses and industries. The system will not include multi-branch synchronization, e-commerce, mobile apps, CRM, loyalty programs, and full-blown accounting. Predictive analytics will just include historical sales forecasting and will not include market trends, competitor analysis, economic, and social trends among others. Predictive analytics will be trained and developed using Risha Pet Supplies' historical



sales and will not include any form of benchmarking from other businesses. The system will also not include multi-currency and different regions, which include different currencies from other nations. Additionally, it will not be multi-lingual and will be developed solely to cater to local operations in the Philippines. The system will also not include RFID, IoT, and other automatic dispensary systems and will include just barcode scanners and receipt printing. Performance analysis will be conducted just during implementation and will no longer include extended periods to test return on investment. Security will also include just basic user log-in and will not include other advanced security features such as blockchain, biometrics, and pen-tests.

Definition of terms

To ensure clarity and consistency throughout this study, the following section presents the key terms used in the research. Since the system developed for Risha Pet Supplies involves a combination of technical processes, data-driven methods, and inventory management concepts, each term is defined based on how it is applied within the study or its standard dictionary meaning. These definitions aim to provide readers with a clear understanding of the technical language used, allowing them to follow the discussions, analyses, and findings more effectively.

Automated Reordering. A process that is operated or controlled by automation for the purpose of ordering again (Merriam-Webster, n.d.-y; Merriam-Webster, n.d.-n).

Barcode Scanning. The process of reading a code consisting of printed and variously patterned bars and spaces into computer memory (Merriam-Webster, n.d.-b).



Database Management System. A usually large collection of data organized especially for rapid search and retrieval combined with a set of procedures and techniques for managing an organization or a particular function (Merriam-Webster, n.d.-ad; Oxford Reference, n.d.-a).

Demand Forecasting. The process of calculating or predicting the quantity of a commodity or service wanted at a specified price and time, usually as a result of study and analysis of available pertinent data (Merriam-Webster, n.d.-c; Merriam-Webster, n.d.-d).

Demand Pattern. A reliable sample of traits, acts, tendencies, or other observable characteristics regarding the quantity of a commodity or service wanted at a specified price and time (Merriam-Webster, n.d.-ag; Merriam-Webster, n.d.-d).

Ensemble Learning. In this study, ensemble learning refers to the combination of multiple machine learning algorithms (Random Forest and Gradient Boosting) working together to produce more accurate inventory forecasts than any single algorithm could achieve independently for Risha Pet Supplies.

Expiration Tracking. The process of following the tracks or traces of or observing the moving path of expiration or termination (Merriam-Webster, n.d.-z; Merriam-Webster, n.d.-aa).



Gradient Boosting. Within this research, Gradient Boosting represents a machine learning technique that sequentially builds prediction models, with each new model correcting errors from previous models to improve forecasting accuracy for Risha Pet Supplies.

Historical Sales Data. Factual information of, relating to, or having the character of history used as a basis for reasoning, discussion, or calculation (Merriam-Webster, n.d.-ab; Merriam-Webster, n.d.-ac).

Inventory. An itemized list of current assets or a quantity of goods or materials on hand (Merriam-Webster, n.d.-f).

Inventory Accuracy. For the purposes of this study, inventory accuracy measures the degree of correspondence between recorded stock quantities in the system and actual physical inventory counts at Risha Pet Supplies.

Inventory Management. The conducting or supervising of an itemized list of current assets or a quantity of goods or materials on hand (Merriam-Webster, n.d.-f; Merriam-Webster, n.d.-g).

Machine Learning (ML). The process by which a computer is able to improve its own performance by continuously incorporating new data into an existing statistical model (Merriam-Webster, n.d.-h).



Point-of-Sale (POS). The place at which a retail transaction is carried out or a computerized system used in such places for recording sales and related functions (Merriam-Webster, n.d.-i).

Predictive Analytics. A branch of advanced analytics that makes predictions about future outcomes using historical data combined with statistical modeling, data mining techniques and machine learning (TechTarget, n.d.).

Random Forest. In this study, Random Forest refers to a machine learning algorithm that creates multiple decision trees using different data samples to generate accurate demand forecasts through collective prediction averaging for Risha Pet Supplies inventory.

Real-Time Inventory. The actual time during which something takes place for an itemized list of current assets or a quantity of goods or materials on hand (Merriam-Webster, n.d.-x; Merriam-Webster, n.d.-f).

Reorder Point. Within this research, reorder point indicates the predetermined inventory level that triggers an automatic purchase order to replenish stock before depletion occurs at Risha Pet Supplies.

Safety Stock. For the purposes of this study, safety stock refers to the buffer inventory quantity maintained above regular demand levels to prevent stockouts during unexpected demand increases or supplier delays at Risha Pet Supplies.



Seasonal Demand. The quantity of a commodity or service wanted at a specified price and time that is of, relating to, or varying in occurrence according to the season (Merriam-Webster, n.d.-d; Merriam-Webster, n.d.-j).

Stock Keeping Unit (SKU). A number or string of alphanumeric characters that uniquely identify a product and used to track inventory in retail stores, warehouses, and online retailers (Merriam-Webster, n.d.-p).

Stockout. The depletion of stock on hand of an item offered for sale (Merriam-Webster, n.d.-l).

Supplier Notification. In this study, supplier notification refers to automated alerts sent to product vendors when Risha Pet Supplies requires inventory replenishment, streamlining the procurement process and reducing order delays.

Supply Chain Management. The management of the flow of goods and services from raw material sources to end customers, including procurement, production, inventory management, and distribution (Oxford Reference, n.d.-d).



Chapter 2

Methods

This chapter discusses the research methodology utilized during the development, design, and evaluation of the Intelligent Inventory Management System with Predictive Analytics for Risha Pet Supplies. This includes the explanation of the design, location, population, sample, research instrument, methods of gathering information, as well as the method of analysis to be utilized.

Research Locale

The study is to be conducted at the business establishment of Risha Pet Supplies based at 07 Bagumbong Road, Brgy. 171, Caloocan City, Metro Manila. This business caters to pet-related item sales, which focuses on products intended for dog and cat companions, with only some available products for birds. The present products include pet food, accessories, health-related products, grooming products, and many more.

Risha Pet Supplies is operating daily from 6am-5pm and serves its customers in the surrounding environs. As of now, the business engages in manual handling to control its inventory. This presents a challenge when it comes to the tracking and demand forecasting of the inventory. Risha Pet Supplies presents an excellent avenue for research in the development and testing of an intelligent inventory management system.

**Research Design**

In a bid to systematically develop, test, and assess FETCH, this study proposes to use particular methodologies that will ensure a systematic development process while concurrently testing its efficiency. FETCH shall adopt a developmental research design methodology that combines the Agile SDLC software development life cycle model with a descriptive-evaluative design.

Agile Software Development Life Cycle (SDLC). In FETCH, the proposed project will adopt a developmental research design with a descriptive-evaluative method involving the Agile Software Development Life Cycle (SDLC). The former will be concerned with the development phase of the Intelligent Inventory Management System, where repeated stages of software development will be undertaken in the design of the system. The study will adopt the Agile SDLC process due to its ability to facilitate continued software development and the inclusion of feedback from stakeholders on the software system design. This is unlike the Waterfall model that involves the sequential process of software system development. The need for adaptability in software system requirements cannot be over-estimated, especially where interactions between the software developers or design teams and software end users may result in an altered software system or new requirements. The process of development is to be done using Agile methodologies such as sprint planning, coordination, sprint reviews, and retrospectives. The descriptive-evaluative component of the research aims to use quantitative approaches in order to obtain data that can reveal the effectiveness of the



system's usability, as well as user satisfaction, using a questionnaire after the hands-on evaluation of the system. It provides a systematic approach to building the system, as well as a way of objectively determining the effectiveness of the system.

Figure 2

Agile Development Model



FETCH would be developed based on the following process:

Planning. The team will leverage this stage to set objectives of FETCH as well as all the requirements needed. These requirements cover server as well as database requirements, as well as requirements of software including Visual Studio Code, MySQL Workbench, as well as Python environment. The team aims at finding all the requirements needed in order to implement functionalities of inventory management, barcode scanner functionality, autoordering functionality, demand forecasting functionality, as well as expiration dates.



Design. The study will comprise the design of the system's interfaces and architecture. The design will be aimed at being user-friendly so that Risha Pet Supplies personnel can use it to control and track the stock levels and also for carrying out transactions. The designs for the dashboard of the management and administrators will entail the use of predictive analytics and notifications.

Development. The system will be developed based on the design that has been accepted. This will entail coding the front end of the system with HTML, CSS, JavaScript, coding of the back end with Java Spring Boot and MySQL databases, as well as coding of machine learning models with Python. The system will entail functionalities such as inventory management, barcode scanner, auto reordering, inventory forecasting with the use of the Random Forest algorithm and Gradient Boosting algorithm.

Testing. Quality assurance and User Acceptance Testing (UAT) will also be conducted to validate if the system works as it should. Technical professionals will test the functionality, reliability, and compatibility of the system, and users at Risha Pet Supplies will test the effectiveness and usability of the system. Tools like UAT testing forms and Computer System Usability Questionnaire (CSUQ) would be used.

Deployment. The system is to be deployed at Risha Pet Supplies, where it will be accessible to staff and management for implementation. This stage will ensure that it is fully integrated with all hardware, software, and barcode scanners, while still incorporating real-time inventory management and replenishment.



Review. The system's efficiency and effectiveness will be measured as feedback will be obtained from stakeholders and the system's users on the system's performance and level of satisfaction with the system.

Respondents of the study

The study will select respondents through purposive sampling to ensure that participants have relevant experience in inventory operations, system usage, technical development, and supply chain processes connected to the FETCH: Inventory Management System with Predictive Analytics. This method supports the developmental and descriptive-evaluative research design by helping researchers gather reliable and practical feedback on the system's usability, technical performance, and effectiveness in real-world situations. The chosen respondents will take part in interviews, system testing, User Acceptance Testing (UAT), and usability evaluations using the Computer System Usability Questionnaire (CSUQ).

The respondents will include one (1) owner and one (1) staff member from Risha Pet Supplies as the main system users. Additionally, five to ten (5-10) suppliers will evaluate the automated reordering and notification features. Five to eight (5-8) technical experts will assess the system's architecture, functionality, and the accuracy of predictive analytics. Lastly, five to ten (5-10) owners or managers of other small pet shops will examine the system's scalability and relevance in the industry. These groups will evaluate FETCH from operational, technical, business, and supply chain perspectives. This



approach makes the assessment thorough and fits the study's research methodology.

Research Instrument

To comprehensively assess FETCH's functionality and usability, the researchers will employ multiple data-gathering instruments. These tools will enable systematic evaluation of system performance, user satisfaction, and alignment with the operational requirements of Risha Pet Supplies.

Documentary Analysis. The researchers will undertake a literature analysis of research works related to the development of inventory management systems, predictive analytics, machine learning technologies in the retail industry, or automated reordering processes to use in building the FETCH system. The sources will cover local literature as well as worldwide research works for demand forecasting methods, ensemble learning approaches, real-time inventory monitoring, or point-of-sale integration. The sources of documentation shall form the basis for developing the techniques of the new system to enhance inventory accuracy, minimize inventory shortages, or maximize inventory levels of Risha Pet Supplies.

Guide Questions. The researchers will hold interviews with the Risha Pet Supplies workforce, managers, as well as with suppliers as potential users of the system with the aim of creating an understanding of the requirements of the system. The team will focus on creating an understanding of how they can use the system or make it useful as it relates to inventory management functionality.



User Acceptance Testing (UAT). Test case documents will be created by the researchers to ensure the FETCH system is functioning as expected. Users from the Risha Pet Supplies will be asked to carry out various activities such as entering a new product entry for sales, managing sales transaction entries, scanning bar codes for products, checking demand predictions, auto-generating purchase orders, and tracking the levels of the current available stock using the system's dashboard. Test activities will also focus on activities related to the system's reliability and functionality between the front-end and back-end built using Java Spring Boot and Python-based machine learning codes.

Computer System Usability Questionnaire (CSUQ). In this respect, the researchers will be using the "Yes"/ "No" dichotomous scale to gauge system usability, the quality of the interface, and user satisfaction. This will further enable staff and managers to give candid feedback about how easy FETCH will be to navigate, how clear the predictive analytics visualizations are, and their general feeling of confidence in using the system to execute everyday inventory management tasks.

Data Gathering Procedure

The data gathering will be facilitated by the researchers in a structured, multi-phased approach, using both quantitative records and qualitative feedback from the management of Risha Pet Supplies. First, the researchers will arrange with the owners of the store to retrieve historical sales data, inventory logs, and supplier invoices from the last three (3) to twelve (12) months. This raw data will be collected through a data extraction template and will be the main input for the training of the ensemble machine



learning models, specifically the Random Forest algorithm. Upon analyzing these historical records, the researchers predict that they will determine sales patterns, such as seasonal fluctuations and spikes in demand on weekends, which are crucial for the predictive functions of the system.

When it comes to the planning or designing cycle of the development life cycle, semi-structured interviews, as well as observations, among the store's employees, will be conducted to ascertain the functional requirements. By using an interview guide as the research instrument, the researchers will encode the deficiencies in the store's existing manual spreadsheet process and the "intuition-based" reordering process. The understanding developed this way aims to ensure that the software interface is user-friendly, based on actual users, like the use of larger elements of the interface for the use of tablets, as determined in the designing cycle.

The researchers will finally collect evaluative data through iterative unit testing and user acceptance sessions. In unit testing, the researchers will utilize test cases to track the system's accuracy in terms of stock level forecasting and the issuance of reorder alerts. As functional modules are completed, the assumed respondents-the staff at Risha Pet Supplies-will be asked to interact with the prototypes. Their feedback will be solicited through standardized evaluation forms that will question them about the usability of the system and the reliability of the AI-driven insights. This iterative collection of data will ensure that the final intelligent inventory management system actually minimizes costs and prevents stockouts as intended by the study.



Statistical Treatment

The section below discusses the statistical techniques and tools to be utilized in the analysis of the data obtained during User Acceptance Testing (UAT) and the Computer System Usability Questionnaire (CSUQ). The techniques will test for usability, functionality, and effectiveness of the system being implemented.

Percentage-Frequency Distribution. A frequency distribution is to be employed to organize all the data that has been collected from the respondents. In percentage-frequency distribution, the total number of responses made on each criterion is represented, giving a visual idea about the distribution of feedback received.

Dichotomous Scale. The study will use a scale of dichotomous (YES/NO response options) in measuring the usability and acceptability level of the system. This is to ensure that the study has a flat response on the acceptability of certain features of the predictive analytics and inventory modules.

This percentage will be calculated for each item and each subscale, and it will also be calculated for system evaluation.

Table 1

Rating for Level of Usability (CSUQ)

In order to give a qualitative aspect to the obtained data, a comparison between the obtained results and a certain benchmark will be carried out. This helps to establish if



the Inventory Management System is ready to be put to use or if it still needs development.

Range	Descriptive rating	Interpretation
$\geq 78\%$	Passed (P)	The item/criteria passed the system evaluation
$< 78\%$	Failed (F)	The item/criteria failed the system evaluation

The criteria against which evaluation of the Inventory Management System is conducted are presented below in Table 1. The passing criteria is set at 78% based on a benchmark, and all systems above this point are rated acceptable in terms of predictive and management functionality, while those below this point fall below user requirements.

**Appendix A****CONSENT FORM**

Dear Ma'am/Sir,

We are students of the Bachelor of Science in Computer Science at De La Salle Araneta University, currently conducting a research study entitled:

"FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies"

Our system is now live and accessible for testing. It is designed to improve inventory management efficiency and support data-driven decision-making for pet supply retail operations. The system features predictive analytics using ensemble machine learning algorithms (Random Forest and Gradient Boosting), automated reordering capabilities, barcode scanning integration, real-time inventory tracking, supplier management, and expiration monitoring. The system has undergone multiple phases of development and has been reviewed and approved by our thesis adviser.

We would like to invite you to explore our system and experience its features firsthand. Afterward, we kindly ask that you complete a brief 19-item questionnaire consisting of simple yes-or-no questions regarding the system's usability and functionality.

Your feedback will be highly valuable in improving our system. All responses will remain confidential and will be used strictly for academic purposes.

Thank you very much for your time and support.

In St. La Salle.

Joylinda P. Jugan
Researcher

Juan Carlo M. Reusora
Researcher

CONSENT APPROVAL

Evaluation Name (Optional): _____

Position / Company: _____

Signature: _____



Computer System Usability Questionnaire (CSUQ)

FETCH: Inventory Management System with Predictive Analytics for Risha Pet Supplies

Please rate the usability of the system.

- Try to respond to all the items by putting a check mark (✓) on the respective column of your response.
- For items that are not applicable, use: NA
- Add a comment about an item if your response is No

Overall Reaction to the System	Yes	No	N/A	Remarks If No, what is your comment.
1. Overall, I am satisfied with how easy the system is to use.				
2. The system is easy and simple to use for managing pet supply inventory.				
3. I can complete inventory tasks (stock tracking, barcode scanning, POS transactions) effectively using the system.				
4. The system allows me to complete my work quickly with automated features.				
5. The predictive analytics and automated reordering help me perform my work efficiently.				
6. I feel comfortable when using the system for daily inventory operations.				



7. Learning to use the system's features (barcode scanning, forecasting, reordering) was easy.				
8. I became productive quickly when using the system.				
9. The system provides clear alerts (low stock, expiration warnings, reorder notifications) that help me take action.				
10. When I make mistakes, I can recover easily and continue using the system.				
11. The system provides clear and helpful information (demand forecasts, inventory reports, supplier notifications).				
12. It is easy to find inventory information, sales data, and forecasting reports I need in the system.				
13. The demand forecasting results and inventory analytics are easy to understand.				
14. The predictive analytics and automated alerts help me make better stocking decisions.				
15. The dashboard and inventory information on the screen are well organized and clear.				
16. The system's interface is pleasant to look at.				



17. I like using the system's interface for inventory management tasks.				
18. The system has all the features I expect (predictive analytics, automated reordering, barcode scanning, supplier management, expiration tracking).				
19. Overall, I am satisfied with the system.				



Appendix B

UNIT TESTING PLAN (User Acceptance Testing)

Project: FETCH: Inventory Management with Predictive Analytics for Risha Pet Supplies

Date: _____

Tested By: _____

1. Completeness Check

Test ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status	Remarks
CC-001	Inventory Tracking	Add product with missing required fields	Refer to UAT procedure	System behaves as expected			
CC-002	Inventory Tracking	Add product with missing SKU and Category	Refer to UAT procedure	System behaves as expected			
CC-003	Automated Reordering	Set reorder settings without supplier	Refer to UAT procedure	System behaves as expected			



CC-004	Point of Sale	Process transaction without payment method	Refer to UAT procedure	System behaves as expected			
CC-005	User Management	Create user account with missing role	Refer to UAT procedure	System behaves as expected			

2. Consistency Check

Test ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status	Remarks
CS-001	Inventory & POS	Verify inventory update after sale	Refer to UAT procedure	System behaves as expected			
CS-002	Inventory & Database	Verify inventory consistency across views	Refer to UAT procedure	System behaves as expected			
CS-003	Reordering & Inventory	Verify stock level after PO received	Refer to UAT procedure	System behaves as expected			
CS-004	POS & Sales Reports	Verify daily sales	Refer to UAT procedure	System behaves			



		report accuracy		as expected			
CS-005	Expiration & Inventory	Verify expired product handling	Refer to UAT procedure	System behaves as expected			
CS-006	User Management & Login	Verify role-based access consistency	Refer to UAT procedure	System behaves as expected			
CS-007	Refund & Inventory	Verify inventory restoration after refund	Refer to UAT procedure	System behaves as expected			

3. Format Check

Test ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status	Remarks
FC-001	Inventory Tracking	Product SKU format validation	Refer to UAT procedure	System behaves as expected			
FC-002	Inventory Tracking	Price format validation	Refer to UAT procedure	System behaves as expected			
FC-003	Supplier Management	Email and phone validation	Refer to UAT procedure	System behaves as expected			



FC-004	Expiration Monitoring	Date format validation	Refer to UAT procedure	System behaves as expected			
FC-005	User Management	Username and password validation	Refer to UAT procedure	System behaves as expected			

4. Range Check

Test ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status	Remarks
RC-001	Inventory Tracking	Product quantity negative and zero values	Refer to UAT procedure	System behaves as expected			
RC-002	Inventory Tracking	Product quantity maximum limit	Refer to UAT procedure	System behaves as expected			
RC-003	Inventory Tracking	Product price negative and zero values	Refer to UAT procedure	System behaves as expected			
RC-004	Automated Reordering	Reorder point validation	Refer to UAT procedure	System behaves as expected			



RC-005	Point of Sale	Discount validation	Refer to UAT procedure	System behaves as expected			
RC-006	Expiration Monitoring	Expiration date range validation	Refer to UAT procedure	System behaves as expected			

5. Functional Check

Test ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status	Remarks
FN-001	Inventory Tracking	Add, update, and delete product	Refer to UAT procedure	System behaves as expected			
FN-002	Inventory Tracking	Search and filter products	Refer to UAT procedure	System behaves as expected			
FN-003	Demand Forecasting	Generate and export forecast	Refer to UAT procedure	System behaves as expected			
FN-004	Automated Reordering	Complete reorder workflow	Refer to UAT procedure	System behaves as expected			



FN-005	Point of Sale	Process sale with barcode and discount	Refer to UAT procedure	System behaves as expected			
FN-006	Point of Sale	Process refund and generate report	Refer to UAT procedure	System behaves as expected			
FN-007	Supplier Management	Add and manage suppliers	Refer to UAT procedure	System behaves as expected			
FN-008	Expiration Monitoring	Track and alert on expiring products	Refer to UAT procedure	System behaves as expected			

6. Database Check

Test ID	Module	Test Scenario	Test Steps	Expected Result	Actual Result	Status	Remarks
DB-001	Inventory Tracking	CRUD operations	Refer to UAT procedure	System behaves as expected			
DB-002	Inventory Tracking	Database constraints	Refer to UAT procedure	System behaves as expected			



DB-003	Point of Sale	Transaction atomicity and recording	Refer to UAT procedure	System behaves as expected			
DB-004	Point of Sale	Cascade operations	Refer to UAT procedure	System behaves as expected			
DB-005	Automated Reordering	Purchase order management	Refer to UAT procedure	System behaves as expected			
DB-006	Supplier Management	Supplier records and relationships	Refer to UAT procedure	System behaves as expected			
DB-007	User Management	User authentication	Refer to UAT procedure	System behaves as expected			

Status Legend

Pass – Test case executed successfully and met expected results

Fail – Test case did not meet expected results

N/A – Test case not applicable

Blocked – Test case cannot be executed due to dependency issues